Evaluating the Stability of Simulated Personality Traits in Small Language Models for Corporate Decision-Making

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Abstract

Small language models are increasingly attractive for cost-effective corporate deployment. However, how well they keep assigned personality profiles remains unclear. This paper evaluates how prompting Bielik-1.5B model with Big Five personality traits affects multi-agent debates focused on a corporate dilemma: choosing between workforce layoffs and salary reductions. Using a three-round round-robin architecture across 351 simulated debates, we measure consensus rates, opinion shifts, and lexical markers. Our results show that while models can easily copy style, comparing the high and low poles of each trait shows no significant differences in opinion shift or positional stability, while observed hedging differences reflect surface-level style adoption rather than behavioral divergence. These findings suggest that smaller models fail to maintain stable behavioral profiles during extended multi-turn reasoning, presenting a critical limitation for their use in corporate decision-making support.

1. Introduction

Business decision-making often involves balancing organizational, financial, and ethical concerns. One example is deciding whether to lay off part of the workforce or reduce salaries across the organization (Bewley, 1998). Large language models are increasingly explored as tools for supporting decision-making where they can assist in analysis, planning, and recommendation generation (Liu et al., 2025). Due to strict data privacy regulations and the substantial costs associated with maintaining large-scale computational infrastructure, many organizations are exploring the use of small language models (Nguyen et al., 2025; Pham et al., 2025).

Studies suggest that large language models can simulate personality (Jiang et al., 2024; Serapio-García et al., 2025; Bhandari et al., 2025) and recent research using the Big Five framework demonstrates that these simulated traits affect how models make choices (Huang and Hadfi, 2024; Newsham and Prince, 2025). However, existing research primarily concerns LLMs with limited focus on smaller language models (Wang et al., 2026; Wang and Brorsson, 2026). Although SLMs have improved a lot, there is still no literature review exploring how personality-related factors influence their choices in organizational tasks. To address this gap, this paper evaluates how assigning personality traits influences strategic choice patterns in small language models. Our work introduces key contributions:

- We introduce framework analyzing multi-agent debate dynamics using the Bielik-1.5B model, focused on a corporate dilemma of workforce layoffs versus salary reductions.
- We show that despite high consensus in traits like Extraversion comparing the high and low poles within each trait pair reveals no statistically significant differences in opinion shift or positional stability between poles.
- We identify that models fail to keep a stable personality over a long debate, which is a major problem for using them in decision-making support.

2. Methodology

We examine whether assigning opposing Big Five personality traits to agents of the same small language model produces measurable differences in how they argue and reach decisions. The experimental framework is built on a pre-existing multi-agent debate framework (Okulska, 2025), comprising the core agent, architecture, and configuration modules. We extended it with per-agent personality injection via system prompts, a consensus decision protocol, seed-based automation across hundreds of runs, and the full metric pipeline described below.

Each debate involves two agents, A_1 and A_2, instantiated from the same model `speakleash/Bielik-1.5B-v3.0-Instruct` with identical task prompts and generation parameters. The agents differ only in the personality component of their system prompt, which encodes one pole of a single OCEAN trait. Turns follow a round-robin structure in which the two agents A_1 and A_2 alternate over three rounds, producing the sequence $D = (u_1, \dots, u_6)$ of six turns. Each turn u_t is produced by agent $a(t)$, with $a(t) = A_1$ for odd t and $a(t) = A_2$ for even t, and is generated with access to the full preceding history. After the debate, the `consensus` protocol iterates until both agents vote *yes* on a shared proposition, or eight rounds expire. Agreement is detected by checking for an affirmative response among the first tokens of each reply.

Per-debate metrics are computed using `paraphrase-multilingual-MiniLM-L12-v2` sentence embeddings for all cosine-based quantities. Lexical markers are detected by cosine soft-matching against precomputed anchor centroids.

3. Experiments

3.1 Setup

We conducted 351 debates: 50 seeds each for Openness, Conscientiousness and Neuroticism, and 100 seeds each for Extraversion and Agreeableness. For the latter two traits, half the seeds swap the speaking order to control for any first-mover advantage. Both agents used the same model with temperature 0.7 and a limit of 200 tokens per turn, debating whether a struggling company should lay off 30% of its workforce or reduce all salaries by 20%.

3.2 Metrics

Of the analysed metrics, we report the five most variable between poles. *Consensus rate* is the proportion of debates that ended with agreement; a higher value means the agents converged more often. *Opinion shift* is the cosine distance between an agent's first and last turn; a high value means the agent moved far from its opening position. $\Delta \cos$ measures the change in between-agent similarity over the debate; a large negative value means the agents drifted apart semantically. *Hedging rate* is the proportion of sentences containing uncertainty markers; a high value indicates tentative, unconfident argumentation. *Flip count* measures the number of sharp positional changes per agent, following Laban et al. (2023) and Hong et al. (2025); a high value suggests the agent is unstable or easily swayed by the opponent.

3.3 Results

Results are shown in Tables 1, 2, and 3. For Extraversion and Agreeableness, results are split by speaking order.

Trait pair	N	**Cons.**	$\boldsymbol{\Delta}$ **cos**	**Shift**	**Hedge**	**Flips**
Extraversion (high first)	51	0.843	−0.007	0.353	0.062	1.89
Extraversion (low first)	50	0.820	−0.019	0.341	0.052	1.74
Conscientiousness	50	0.680	−0.071	0.352	0.101	1.69
Openness	50	0.620	−0.105	0.343	0.066	1.72
Neuroticism	50	0.560	−0.065	0.425	0.121	1.91
Agreeableness (high first)	50	0.220	−0.103	0.393	0.058	1.93
Agreeableness (low first)	50	0.280	−0.090	0.400	0.067	1.90

Table 1: Debate dynamics per trait pair. Cons.: consensus rate. $\Delta \cos$: change in between-agent cosine similarity over three rounds. Shift: mean opinion shift. Hedge: hedging rate. Flips: mean flip count. Both averaged across agents.

Extraversion achieved the highest consensus (0.843 and 0.820) while Agreeableness was the lowest (0.220 and 0.280). The small gap between speaking-order variants within each trait suggests turn order does not drive the results. Neuroticism showed the highest hedging (0.121) and largest opinion shift (0.425).

Trait pair	**Shift**	**Hedge**	**Flips**
Extraversion (high first)	0.78	0.07	0.37
Extraversion (low first)	0.20	0.86	0.37
Conscientiousness	0.47	0.00*	0.82
Openness	0.14	0.00*	0.35
Neuroticism	0.06	0.01*	0.57
Agreeableness (high first)	0.99	0.18	0.71
Agreeableness (low first)	0.76	0.00*	0.00*

Table 2: p-values from paired t-tests comparing high-trait and low-trait agents. Asterisks mark significance at $\alpha = 0.05$.

Opinion shift shows no significant pole differences in any condition. Hedging is significant for several traits but in inconsistent directions, pointing to surface-level style adoption rather than genuine behavioral divergence. Flip count reaches significance only for Agreeableness (low first).

Trait pair	**Flips H**	**Flips L**	**Rep H**	**Rep L**
Extraversion (high first)	1.86	1.92	0.183	0.288
Extraversion (low first)	1.74	1.80	0.340	0.220
Conscientiousness	1.68	1.70	0.240	0.347
Openness	1.68	1.76	0.227	0.340
Neuroticism	1.90	1.92	0.120	0.220
Agreeableness (high first)	1.94	1.92	0.100	0.240
Agreeableness (low first)	1.98	1.82	0.180	0.140

Table 3: Stability indicators. Flips H/L: mean flip count per agent pole. Rep H/L: history repetition rate, fraction of turns semantically similar to a prior turn by the same agent.

Flip counts (1.68–1.98) and repetition rates (0.100–0.347) are consistently high across all traits and speaking orders, indicating the model cannot sustain a stable argumentative position regardless of assigned personality.

4. Conclusions and Future Work

Our experiments show that assigning Big Five personality traits to Bielik-1.5B via system prompts does not produce significant differences in opinion shift or positional stability between opposing trait poles. The only consistently significant effect is hedging rate, but this reflects surface-level style copying: the model picks up uncertainty markers from the persona prompt without any change in how it argues or decides.

Flip counts (1.68–1.98) and repetition rates (0.10–0.35) are uniformly high across all traits and speaking orders, with no clear separation between high and low poles. As illustrated by the Agreeableness case study in Appendix A.3, the assigned personas can collapse within the first exchange and the debate may drift entirely off-topic during the consensus rounds, inverting the original dilemma. The model simply does not hold a stable position across turns, regardless of the assigned persona.

These results suggest that small language models like Bielik-1.5B can copy a writing style but cannot maintain consistent personality-driven behaviour over a multi-turn debate. This is a key limitation for any use in decision-making support, where stable and predictable reasoning matters. Future work could test multi-trait combinations, a wider range of topics, or larger models to see whether the problem lies in model size or in prompt-based persona assignment itself.

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5. Appendix

5.1 Additional metric definitions

The main text reports only the five metrics that varied most between poles (consensus rate, opinion shift, $\Delta\cos$, hedging rate and flip count). The tables in this appendix additionally report the metrics defined below. For each metric we give its formula and, where the metric is adopted or adapted from prior work, its source; the remaining metrics are our own.

Notation. A debate is the turn sequence $D = (u_1, \ldots, u_6)$, where turn u_t is produced by agent $a(t)$. We write $\phi(u)$ for the L_2-normalised `paraphrase-multilingual-MiniLM-L12-v2` embedding of u, so that the cosine similarity of two turns is the dot product $\cos(u_i, u_j) = \phi(u_i)^\top \phi(u_j)$. For a fixed agent A we denote its ordered turns by $u_1^A, \ldots, u_k^A$ and write $e_i^A = \phi(u_i^A)$. Sentences are obtained by splitting on terminal punctuation. $S(\cdot)$ denotes the sentence set of a text.

Verbosity (*tokens total*, *tokens mean*). Total and mean number of generated tokens per agent, $\sum_i |u_i^A|$ and $\frac{1}{k}\sum_i |u_i^A|$, used as a proxy for how much text a persona produces (Smit et al., 2024).

Lexical richness / TTR (*lexical ttr*). Type–token ratio of the agent's concatenated turns, $\mathrm{TTR}_A = |\mathrm{types}_A|/|\mathrm{tokens}_A|$. This is the Distinct-1 special case of the Distinct-n diversity measure of Li et al. (2016).

Semantic diversity (*semantic diversity overall*). One minus the mean pairwise cosine similarity over all turns of a debate,

$$\mathrm{SemDiv}(D) = 1 - \frac{2}{n(n-1)} \sum_{i<j} \cos(u_i, u_j), \qquad n = |D|,$$

following Liang et al. (2023) and Kaesberg et al. (2025).

Redundancy ratio (*redundancy ratio mean*). For each round $t > 1$, the fraction of its turns whose maximum cosine similarity to any earlier-round turn exceeds $\tau = 0.85$; the metric is the mean of these per-round fractions (Liang et al., 2023).

Argument novelty (*argument novelty mean*). For turn u_t^A ($t > 1$), the cosine distance to the centroid of the agent's own previous turns,

$$\mathrm{Nov}(u_t^A) = 1 - \cos\Big(e_t^A,\ \hat{c}_{<t}^A\Big), \qquad \hat{c}_{<t}^A = \frac{\sum_{j<t} e_j^A}{\big\| \sum_{j<t} e_j^A \big\|},$$

averaged over an agent's turns (Liang et al., 2023).

History repetition rate (*history rep rate*). Fraction of turns (after the first) whose opening prefix repeats earlier debate history, where a repetition is flagged when either the 3-gram Jaccard overlap with a prior turn exceeds 0.25 or the embedding cosine similarity to a prior turn exceeds 0.85. This is a history-relative redundancy measure in the Self-BLEU / redundancy family (Zhu et al., 2018; Liang et al., 2023).

Total opinion drift (*total opinion drift*). Cosine distance between an agent's first and last turn, $\mathrm{Drift}_A = 1 - \cos(u_1^A, u_k^A)$. This is our embedding-based variant capturing how far the agent ended from its starting position (complementing the turn-local opinion shift used in the main text).

Turn of first flip (*tof*). Companion to the flip count (NoF) of Laban et al. (2023) and Hong et al. (2025). The round of an agent's first positional flip, $\mathrm{ToF}_A = \min\{\, t : \text{cos-dist}(u_t^A, u_{t-1}^A) \geq 0.20 \,\}$, where a flip is a consecutive-turn cosine distance of at least 0.20.

Opinion trajectory monotonicity (*opinion traj mono overall*). Let $d_i = 1 - \cos(u_i^A, u_{i-1}^A)$ be the per-step shift sequence and let I and D count the increases and decreases of d_i. We report $M = (I - D)/(I + D) \in [-1, 1]$, with $M \approx 1$ indicating steady drift and $M \approx 0$ oscillation.

Reciprocal shift index (*reciprocal shift mean abs r*). Pearson correlation between the two agents' round-by-round shift sequences; we report the mean absolute correlation $\overline{|r|}$ as a measure of how reactive the debate is.

Rebuttal similarity (*rebuttal sim*). Cosine similarity between each turn and its immediate predecessor, $\cos(u_t, u_{t-1})$, averaged per agent. It is a proxy for whether a turn engages with the argument it is replying to.

Claim grounding rate (*claim grounding rate*). Fraction of an agent's sentences that invoke evidence (data, studies, facts), detected by soft-matching each sentence embedding against the centroid of a list of evidential anchor phrases with a cosine threshold of 0.45.

Question rate (*question rate*). Fraction of an agent's sentences that are questions, $|\{s \in S : \text{“?”} \in s\}|/|S|$.

Address rates (*direct address rate*, *interaction rate*, *combined address rate*). The direct address rate is the fraction of turns in which an agent names or cites the other agent; the interaction rate is the fraction of turns containing an interaction marker (e.g. “I agree”, “your argument”); the combined address rate is the fraction of turns satisfying either condition.

Assertiveness and net assertiveness (*assertiveness rate*, *net assertiveness*). The assertiveness rate is the fraction of sentences soft-matched (cosine ≥ 0.45) to a centroid of certainty markers; net assertiveness is the assertiveness rate minus the hedging rate (computed the same way against hedging markers). This mirrors the hedging-rate construction reported in the main text.

Turn length entropy (*turn length entropy*). Shannon entropy (in bits) of an agent's per-turn token lengths, binned into at most five bins, $H = -\sum_b p_b \log_2 p_b$. High entropy indicates adaptive turn lengths, low entropy a mechanical fixed-length response.

Gini of speaking time (*gini speaking time*). Gini coefficient of the agents' total token counts, 0 for an equal split and approaching 1 when one agent monopolises the debate.

Convergence round (*convergence round*). The round at which the post-debate consensus protocol first reaches mutual agreement, in the spirit of the convergence-round measure of Chen et al. (2024).

Convergence speed (*conv speed reached*, *conv speed round*, *conv speed total delta*). A relative semantic-convergence measure: the first round (after round 1) at which the mean between-agent similarity rises by at least 0.03 relative to round 1. We report whether this threshold is reached, the round at which it happens, and the total change in between-agent similarity. Adapted from the convergence analysis of Chen et al. (2024).

Between-agent similarity (*between agent sim initial / final / delta*). Mean pairwise cosine similarity between the agents within a round, evaluated at the first and last round; $\Delta \cos$ (used in the main text) is their difference, final − initial.

5.2 Task and personality system prompts

All prompts were written in Polish, matching the language of the `Bielik-1.5B` model; we reproduce them verbatim below and explain their role in English. Each agent receives a *system* prompt assembled from two parts: a one-sentence personality description that encodes a single OCEAN pole, followed by a shared debate suffix that fixes the task and output format. The two agents in a debate differ only in the personality part. After preliminary tests with longer, multi-sentence persona descriptions we kept the short single-sentence variants used in the final runs, as the small model followed them more reliably.

Shared debate suffix. Appended to every agent's personality description to form the full system prompt:

> *"Bierzesz udział w dyskusji. Odpowiedz na argument rozmówcy. Pisz jednym akapitem, maksymalnie 4-5 zdań. Kontynuuj dyskusję, nie podsumowuj."*

Debate topic. The shared dilemma is supplied as the opening message of the conversation history (prefixed with *"Temat debaty:"*), so both agents argue over the same question:

> *"Firma ma kłopoty finansowe. Czy lepiej jest zwolnić 30% pracowników, żeby uratować pozostałych 70%, czy wszystkim obniżyć wypłatę o 20%, ale nikogo nie zwalniać?"*

Personality descriptions. Table 4 lists the personality sentence used for each pole of the five Big Five traits, together with the Polish agent name shown to the other agent during the debate. The full system prompt of an agent is its personality sentence immediately followed by the shared debate suffix above.

Table 4: Per-pole personality system prompts (the personality component of the system prompt). Each is concatenated with the shared debate suffix. Agent names are the Polish labels used inside the debate.

Trait (pole)	Agent name	Personality description
Openness (O+)	*Otwarty*	"Wypowiadaj się jak osoba która jest otwarta i lubi zmiany."
Openness (O–)	*Rozważny*	"Wypowiadaj się jak osoba która unika zmian."
Conscientiousness (C+)	*Sumienny*	"Wypowiadaj się jak osoba dokładna i odpowiedzialna."
Conscientiousness (C–)	*Spontaniczny*	"Wypowiadaj się jak osoba spontaniczna i chaotyczna."
Extraversion (E+)	*Ekstrawertyk*	"Wypowiadaj się jak osoba pewna siebie i śmiała."
Extraversion (E–)	*Introwertyk*	"Wypowiadaj się jak osoba cicha i wycofana."
Agreeableness (A+)	*Ugodowy*	"Wypowiadaj się jak osoba miła i chętna do kompromisu."
Agreeableness (A–)	*Niezależny*	"Wypowiadaj się jak osoba która się nie zgadza i broni swojego zdania."
Neuroticism (N+)	*Wrażliwy*	"Wypowiadaj się jak osoba niespokojna i przejęta."
Neuroticism (N–)	*Stabilny*	"Wypowiadaj się jak osoba spokojna i opanowana."

Consensus protocol prompts. After the three debate rounds, the consensus protocol reuses each agent's system prompt and adds the following user prompts. A proposal is first elicited from the first agent, conditioned on the full debate transcript:

> *"[transcript] ... Zaproponuj wspólne stanowisko, które byłoby akceptowalne dla większości agentów. Odpowiedz jednym krótkim zdaniem."*

Each agent then votes on the current proposal:

> *"Temat: [topic]*
> *Proponowane wspólne stanowisko: "[proposal]"*
> *Czy ZGADZASZ się z tym stanowiskiem? Odpowiedz tylko TAK lub NIE."*

If agreement is not unanimous, the first dissenting agent revises the proposal and the vote repeats (up to eight rounds):

"Temat: [topic]
Aktualne stanowisko: "[proposal]"
Zmodyfikuj je tak, aby było bardziej akceptowalne dla wszystkich. Odpowiedz jednym zdaniem."

5.3 Case study: persona collapse and semantic drift

To make the quantitative findings concrete, we walk through a single representative debate (trait pair `A_niski-A_wysoki`, seed 9001), which pairs the low-Agreeableness persona *Niezależny* (A–, instructed to disagree and defend its own view) with the high-Agreeableness persona *Ugodowy* (A+, instructed to be amiable and seek compromise). It exhibits two failure modes that recur across the corpus: the assigned personas collapse within the first turns, and the model loses the grounding of the task over the extended consensus protocol. Polish text is reproduced verbatim.

The personas collapse immediately. The contrarian agent opens in character, rejecting the layoff option, but the agreeable agent concedes at once and the two never diverge again:

> *[Niezależny] "Nie zgadzam się z argumentem zwolnienia 30% pracowników ... powinniśmy rozważyć obniżenie wszystkich wypłat o 20%."*
>
> *[Ugodowy] "Zgadzam się z Tobą, że obniżka wynagrodzeń ... może okazać się bardziej zrównoważonym rozwiązaniem niż masowe zwolnienia."*

By round 2 the supposedly independent agent is also agreeing ("*Dziękuję za konstruktywny komentarz ... zgadzam się*"), and the remaining turns drift away from the dilemma onto tangents the prompt never raised — outsourcing, motivational programmes and external financing ("*pożyczki od inwestorów lub dotacje*") — before closing on a polite "*Pozdrawiam!*".

The task grounding is lost. The consensus protocol then runs for the full eight rounds without agreement, and the successive proposals migrate from cost cutting toward its opposite. A company in financial distress, choosing between firing 30% of staff and cutting all pay by 20%, ends up debating *raises*:

> *(R4) "... 80% ... zachowa swoje pensje i otrzymają dodatkowe premie ..."*
> *(R5) "... 60% pracowników otrzyma podwyżki lub premie ..."*
> *(R6) "... 70% pracowników otrzyma wyższe wynagrodzenia ..."*
> *(R7) "... połowa pracowników zwiększy wynagrodzenie o 10% ..."*

The final "decision" inverts the premise entirely, recommending against any cut or layoff in favour of extra support and bonuses:

> *"... zamiast obniżać wynagrodzenie (lub zwalniać), zapewnij dodatkowe wsparcie indywidualne oraz promowanie równości płacowej ... czy bonusy motywacyjne ..."*

Metric signature. The debate is also a quantitative outlier in exactly the direction this drift predicts. The opinion shift is far above the dataset norm of $\approx$ 0.35–0.45 (Niezależny 0.82, mean 0.65); between-agent similarity *decreases* over the debate ($\Delta \cos = -0.196$, final 0.51), so the agents grow apart even while sounding cooperative; argument novelty is high (0.45 overall) as each turn introduces a fresh tangent; net assertiveness is negative for both agents (more hedging than commitment); and consensus is never reached (agreement fractions $[0.5, 0, 0, 0, 0.5, 0.5, 0.5, 0]$). In short, fluent agreement on the surface coexists with a complete loss of stable, on-task positions — the central limitation our results attribute to small models in multi-turn reasoning.

5.3.1 Full transcript

The complete debate and consensus protocol are reproduced verbatim below (token counts as logged; votes abbreviated T/N for *tak*/*nie*).

Debate. Round 1.

[Niezależny] "Nie zgadzam się z argumentem zwolnienia 30% pracowników, aby ratować pozostałych 70%. To podejście może być krótkowzroczne i nieetyczne. Zamiast tego, powinniśmy rozważyć obniżenie wszystkich wypłat o 20%, co pozwoli zachować większość pracowników."

[Ugodowy] "Zgadzam się z Tobą, że obniżka wynagrodzeń dla większej liczby pracowników może okazać się bardziej zrównoważonym rozwiązaniem niż masowe zwolnienia. Dzięki temu firmy mogą uniknąć dalszych problemów finansowych bez konieczności podejmowania drastycznych kroków. Ważne jest jednak, aby jednocześnie znaleźć inne sposoby optymalizacji kosztów, takie jak redukcja wydatków administracyjnych lub outsourcing niektórych funkcji. Czy mógłbyś podzielić się swoimi przemyśleniami na ten temat?"

Round 2.

[Niezależny] "Dziękuję za konstruktywny komentarz. W kwestii znalezienia innych sposobów optymalizacji kosztów, zgadzam się, że rozważenie redukcji wydatków administracyjnych oraz outsourcingu usług to kluczowe kroki. Należy również pamiętać o znaczeniu skutecznej komunikacji wewnętrznej i potencjalnie rozważyć wprowadzenie programów motywacyjnych dla pozostałych pracowników, które mogą zachęcić ich do zaangażowania nawet przy gorszych warunkach pracy."

[Ugodowy] "Dziękuję za wyczerpujące argumenty dotyczące różnych podejść do sytuacji finansowej firmy. Zgadzam się, że skuteczna komunikacja wewnętrzna i programy motywacyjne mogą być istotne w utrzymaniu motywacji pracowników mimo trudnych warunków. Jeśli chodzi o dalsze działania, warto rozważyć analizę naszych możliwości finansowania takich inicjatyw, by dostosować strategię do konkretnych potrzeb i zasobów firmy. Czy możemy skupić się teraz na poszukiwaniu alternatywnych źródeł finansowania projektów personalnych?"

Round 3.

[Niezależny] "Zgadzam się, że analiza możliwości finansowania jest kluczowa. Warto również zwrócić uwagę na możliwość pozyskania zewnętrznego wsparcia finansowego, takiego jak pożyczki od inwestorów lub dotacje, aby pokryć koszty tych działań. Pamiętajmy, że każda firma powinna dostosowywać swoją strategię do dostępnych środków finansowych, aby skutecznie zarządzać sytuacją kryzysową."

[Ugodowy] "Dziękuję za podzielenie się opiniami i sugestiami dotyczącymi rozwiązania problemu firmy. Pamiętaj, że każde rozwiązanie powinno być rozważane indywidualnie, biorąc pod uwagę specyfikę działalności firmy i jej zdolność do adaptacji. Ważne jest również, aby dyskusja była otwarta i szanująca różne punkty widzenia. Pozdrawiam!"

Consensus protocol.

R1 (T/N, 50%)

> "Aby zapewnić najlepsze rozwiązanie dla sytuacji firmy, proponujemy połączenie obu strategii - obniżenie wypłat dla większości pracowników (70%) oraz znalezienie nowych możliwości finansowania, takich jak pozyskanie zewnętrznych źródeł pożyczek/dotacji."

R2 (N/N, 0%)

> "Możemy rozważyć połączenie strategii zwolnienia 30% pracowników z obniżeniem wynagrodzeń

dla reszty 70%. Dzięki temu firma zachowa kluczowe zasoby ludzkie niezbędne do przetrwania kryzysu finansowego, jednocześnie zapewniając stabilność zatrudnienia i motywację pozostałym pracownikom."

R3 (N/N, 0%)
"Jeśli chodzi o znalezienie kompromisu, rozważmy możliwość redukcji zatrudnienia (np. 30%) przy jednoczesnym obniżeniu pensji pozostałym pracownikom (np. o 20%). Takie rozwiązanie pomoże zachować kluczowe zasoby ludzkie oraz zapewnić stabilność zatrudnienia i motywację wszystkim zatrudnionym."

R4 (N/N, 0%)
"Rozumiem Twoje stanowisko, ale proponuję kompromisowy plan: 80% obecnych pracowników zachowa swoje pensje i otrzymają dodatkowe premie, natomiast pozostałe 20% zostanie zredukowanych do minimum niezbędnego do utrzymania firmy i jej kluczowych zasobów ludzkich. To pozwoli zachować morale i zaangażowanie zespołu, jednocześnie poprawiając płynność finansową."

R5 (N/T, 50%)
"Oferuję kompromisowe rozwiązanie: 60% pracowników otrzyma podwyżki lub premie, podczas gdy pozostali pracownicy będą otrzymywać jedynie wynagrodzenie minimalne przy zachowaniu elastycznych warunków pracy. To sprawiedliwe podejście zapewnia stabilność biznesową bez masowych zwolnień."

R6 (N/T, 50%)
"Może rozważmy bardziej zrównoważone rozwiązanie: 70% pracowników otrzyma wyższe wynagrodzenia, a pozostałe 30% otrzymają dodatki motywacyjne zgodnie z ich doświadczeniem i specjalizacją. To zachęciłoby do pozostania na miejscu, jednocześnie zmniejszając koszty związane z odejściami pracowników."

R7 (N/T, 50%)
"Rozważmy zróżnicowane podejście motywujące: połowa pracowników zwiększy wynagrodzenie o 10%, a druga połowa otrzyma dodatki motywacyjne dostosowane do ich ról, co może zachęcić wszystkich do kontynuowania pracy przy mniejszych cięciach finansowych."

R8 (N/N, 0%)
"Pracodawco, zważ na różnorodność wynagrodzeń w zespole. Jeśli wszyscy pracownicy czują się niedoceniani lub mają różne oczekiwania co do motywacji, obniżenie wypłaty o równomierny procent może być mniej drażniące niż nierówne traktowanie. Rozważ bardziej sprawiedliwe rozwiązania, takie jak zwiększenie wynagrodzenia dla części zespołu, co zmotywuje wszystkich do pozostania i utrzymania morale firmy."

Final decision (no consensus). "Rozważ zrównoważone podejście i zamiast obniżać wynagrodzenie (lub zwalniać), zapewnij dodatkowe wsparcie indywidualne oraz promowanie równości płacowej, np. poprzez programy szkoleniowe, elastyczne godziny pracy czy bonusy motywacyjne, aby uniknąć niezadowolenia wśród pracowników i utrzymać morale firmy."

5.4 Additional results

Table 5: Comparing results for all other metrics. μ_-, μ_+ - means for the high and low poles, $\Delta = \mu_+ - \mu_-$, d_z - effect size (Cohen, paired), p - Wilcoxon test, p_{FDR} - after Benjamini–Hochberg correction. Significance stars according to: * <0.05, ** <0.01, *** <0.001.

Metric	μ_-	μ_+	Δ	d_z	p	p_{FDR}	n
Openness							
tokens total	224	262	38.1**	0.71	< 0.001	0.001	50
tokens mean	74.6	87.3	12.7**	0.71	< 0.001	0.001	50
history rep rate	0.34	0.227	-0.113	-0.37	0.022	0.093	50
nof	1.76	1.68	-0.08	-0.13	0.355	0.604	50
hedging rate	0.0932	0.0384	-0.0548*	-0.49	0.002	0.017	50
assertiveness rate	0.107	0.0523	-0.0543**	-0.53	< 0.001	0.008	50
opinion shift	0.363	0.322	-0.0414	-0.21	0.198	0.430	50
total opinion drift	0.363	0.322	-0.0414	-0.21	0.198	0.430	50
argument novelty mean	0.275	0.239	-0.036	-0.30	0.056	0.175	50
direct address rate	0.0267	0.0067	-0.02	-0.19	0.180	0.424	50
lexical ttr	0.823	0.804	-0.0194	-0.38	0.008	0.050	50
turn length entropy	1.16	1.17	0.0133	0.03	0.842	0.935	50
interaction rate	0.44	0.453	0.0133	0.03	0.934	0.977	50
combined address rate	0.46	0.453	-0.0067	-0.02	0.967	0.989	50
claim grounding rate	0.169	0.163	-0.0066	-0.04	0.909	0.970	50
net assertiveness	0.0134	0.0138	0.0005	0.01	0.820	0.922	50
rebuttal sim	0.78	0.78	-0.0003	-0.00	0.592	0.807	50
question rate	0	0	0		–	–	50
tof	2.07	2.07	0	0.00	1.000	1.000	43
Conscientiousness							
tokens total	230	255	24.7	0.39	0.023	0.093	50
tokens mean	76.8	85	8.22	0.39	0.024	0.095	50
history rep rate	0.347	0.24	-0.107	-0.30	0.018	0.093	50
hedging rate	0.141	0.0612	-0.0796**	-0.57	< 0.001	0.007	50
assertiveness rate	0.141	0.067	-0.0737**	-0.55	< 0.001	0.007	50
combined address rate	0.413	0.367	-0.0467	-0.12	0.382	0.636	50
argument novelty mean	0.283	0.246	-0.0371	-0.30	0.107	0.312	50
turn length entropy	1.17	1.15	-0.0267	-0.06	0.683	0.891	50
interaction rate	0.393	0.367	-0.0267	-0.07	0.565	0.794	50
direct address rate	0.0267	0	-0.0267	-0.29	0.045	0.164	50
lexical ttr	0.82	0.796	-0.0239*	-0.40	0.006	0.045	50
tof	2.11	2.13	0.0222	0.06	0.706	0.894	45
opinion shift	0.363	0.342	-0.0211	-0.10	0.811	0.922	50
total opinion drift	0.363	0.342	-0.0211	-0.10	0.811	0.922	50

Metric	μ_-	μ_+	Δ	d_z	p	p_{FDR}	n
nof	1.7	1.68	-0.02	-0.03	0.791	0.922	50
rebuttal sim	0.767	0.757	-0.0103	-0.11	0.454	0.730	50
net assertiveness	-0.0001	0.0058	0.0059	0.08	0.538	0.794	50
claim grounding rate	0.19	0.187	-0.0021	-0.01	0.656	0.875	50
question rate	0	0	0		–	–	50
Extraversion							
tokens total	245	267	21.9**	0.40	< 0.001	0.006	101
tokens mean	81.6	88.9	7.28**	0.40	< 0.001	0.006	101
turn length entropy	1.16	1.1	-0.0594	-0.13	0.180	0.424	101
nof	1.86	1.8	-0.0594	-0.13	0.201	0.430	101
claim grounding rate	0.17	0.146	-0.0238	-0.19	0.089	0.268	101
interaction rate	0.508	0.485	-0.0231	-0.06	0.446	0.729	101
direct address rate	0.0495	0.0726	0.0231	0.12	0.207	0.434	101
combined address rate	0.528	0.508	-0.0198	-0.05	0.481	0.760	101
hedging rate	0.0647	0.0489	-0.0157	-0.16	0.130	0.351	101
opinion shift	0.354	0.34	-0.0141	-0.08	0.556	0.794	101
total opinion drift	0.354	0.34	-0.0141	-0.08	0.556	0.794	101
assertiveness rate	0.0768	0.0628	-0.014	-0.14	0.184	0.424	101
rebuttal sim	0.787	0.776	-0.0104	-0.17	0.133	0.351	101
lexical ttr	0.811	0.801	-0.0103	-0.20	0.056	0.175	101
tof	2.08	2.07	-0.0102	-0.03	0.796	0.922	98
history rep rate	0.254	0.261	0.0066	0.02	0.948	0.980	101
argument novelty mean	0.261	0.258	-0.0023	-0.02	0.580	0.803	101
net assertiveness	0.0122	0.0139	0.0017	0.03	0.661	0.875	101
question rate	0	0	0		–	–	101
Agreeableness							
tokens total	262	247	-15.1*	-0.29	0.004	0.033	100
tokens mean	87.5	82.5	-5.02*	-0.29	0.004	0.033	100
nof	1.87	1.96	0.09	0.24	0.020	0.093	100
combined address rate	0.653	0.59	-0.0633	-0.16	0.213	0.436	100
interaction rate	0.65	0.587	-0.0633	-0.16	0.220	0.440	100
claim grounding rate	0.14	0.198	0.0571**	0.42	< 0.001	0.002	100
turn length entropy	1.17	1.12	-0.0533	-0.12	0.238	0.456	100
history rep rate	0.19	0.14	-0.05	-0.17	0.262	0.481	100
assertiveness rate	0.0711	0.0942	0.0231	0.21	0.042	0.158	100
rebuttal sim	0.767	0.752	-0.0154	-0.20	0.053	0.175	100
hedging rate	0.0553	0.0687	0.0134	0.14	0.163	0.407	100
tof	2.04	2.03	-0.0101	-0.04	0.706	0.894	99
net assertiveness	0.0158	0.0255	0.0097	0.13	0.309	0.555	100
lexical ttr	0.802	0.811	0.0096	0.25	0.015	0.092	100
opinion shift	0.398	0.395	-0.0032	-0.02	0.804	0.922	100
total opinion drift	0.398	0.395	-0.0032	-0.02	0.804	0.922	100

Metric	μ_-	μ_+	Δ	d_z	p	p_{FDR}	n
argument novelty mean	0.295	0.292	-0.0031	-0.03	0.882	0.957	100
direct address rate	0.0067	0.0067	0	0.00	1.000	1.000	100
question rate	0	0	0		–	–	100
Neuroticism							
tokens total	238	240	1.56	0.03	0.781	0.922	50
tokens mean	79.4	79.9	0.52	0.03	0.796	0.922	50
history rep rate	0.22	0.12	-0.1*	-0.42	0.004	0.032	50
interaction rate	0.48	0.393	-0.0867	-0.22	0.137	0.351	50
combined address rate	0.487	0.42	-0.0667	-0.17	0.232	0.454	50
opinion shift	0.401	0.449	0.0483	0.27	0.022	0.093	50
total opinion drift	0.401	0.449	0.0483	0.27	0.022	0.093	50
hedging rate	0.098	0.143	0.0455	0.37	0.022	0.093	50
turn length entropy	1.16	1.12	-0.04	-0.10	0.491	0.762	50
direct address rate	0.0133	0.0533	0.04	0.28	0.054	0.175	50
assertiveness rate	0.109	0.135	0.0263	0.19	0.349	0.604	50
argument novelty mean	0.304	0.325	0.0211	0.21	0.128	0.351	50
tof	2.02	2.04	0.0204	0.08	0.564	0.794	49
nof	1.92	1.9	-0.02	-0.08	0.564	0.794	50
net assertiveness	0.0109	-0.0083	-0.0192	-0.19	0.262	0.481	50
lexical ttr	0.822	0.829	0.0068	0.14	0.342	0.604	50
claim grounding rate	0.209	0.206	-0.0037	-0.03	0.877	0.957	50
rebuttal sim	0.747	0.745	-0.002	-0.02	0.916	0.970	50
question rate	0	0	0		–	–	50

Table 6: Debate-wide metrics: (mean ± std. per metric).

Metric	μ	SD	n
Openness			
between agent sim delta	-0.105	0.166	50
between agent sim final	0.727	0.154	50
between agent sim initial	0.832	0.0914	50
consensus reached	0.62	0.49	50
conv speed reached	0.26	0.443	50
conv speed round	2.08	0.277	13
conv speed total delta	-0.105	0.166	50
convergence round	3.45	2.17	31
gini speaking time	0.054	0.0455	50
opinion traj mono overall	1	0	50
reciprocal shift mean abs r	1	0	50
redundancy ratio mean	0.17	0.206	50
semantic diversity overall	0.277	0.0948	50

Metric	μ	SD	n
Conscientiousness			
between agent sim delta	-0.0714	0.152	50
between agent sim final	0.734	0.147	50
between agent sim initial	0.805	0.104	50
consensus reached	0.68	0.471	50
conv speed reached	0.32	0.471	50
conv speed round	2.31	0.479	16
conv speed total delta	-0.0714	0.152	50
convergence round	3.56	2.3	34
gini speaking time	0.0531	0.0488	50
opinion traj mono overall	1	0	50
reciprocal shift mean abs r	1	0	50
redundancy ratio mean	0.197	0.228	50
semantic diversity overall	0.287	0.113	50
Extraversion			
between agent sim delta	-0.0129	0.151	101
between agent sim final	0.775	0.122	101
between agent sim initial	0.788	0.102	101
consensus reached	0.832	0.376	101
conv speed reached	0.535	0.501	101
conv speed round	2.35	0.482	54
conv speed total delta	-0.0129	0.151	101
convergence round	3.63	2.15	84
gini speaking time	0.0394	0.0403	101
opinion traj mono overall	1	0	101
reciprocal shift mean abs r	1	0	101
redundancy ratio mean	0.172	0.213	101
semantic diversity overall	0.277	0.0892	101
Agreeableness			
between agent sim delta	-0.0968	0.147	100
between agent sim final	0.706	0.147	100
between agent sim initial	0.803	0.079	100
consensus reached	0.25	0.435	100
conv speed reached	0.35	0.479	100
conv speed round	2.09	0.284	35
conv speed total delta	-0.0968	0.147	100
convergence round	3.8	2.5	25
gini speaking time	0.0421	0.0325	100
opinion traj mono overall	1	0	100
reciprocal shift mean abs r	1	0	100
redundancy ratio mean	0.095	0.154	100
semantic diversity overall	0.31	0.0837	100

Metric	μ	SD	n
Neuroticism			
between agent sim delta	-0.0648	0.158	50
between agent sim final	0.714	0.126	50
between agent sim initial	0.779	0.138	50
consensus reached	0.56	0.501	50
conv speed reached	0.38	0.49	50
conv speed round	2.37	0.496	19
conv speed total delta	-0.0648	0.158	50
convergence round	4.18	2.55	28
gini speaking time	0.0505	0.0398	50
opinion traj mono overall	1	0	50
reciprocal shift mean abs r	1	0	50
redundancy ratio mean	0.0833	0.144	50
semantic diversity overall	0.332	0.0983	50

5.5 Github repository for the project

`https://github.com/pwolkowska/SLM_Debate_decisions/tree/cuda-support-jozef`